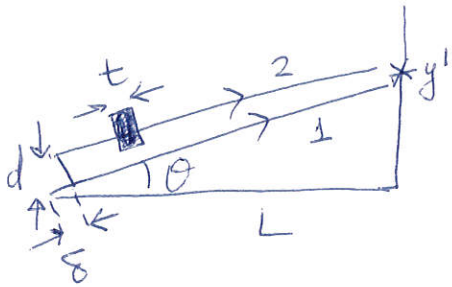
Midterm 1

P1



$\delta = d \sin\theta$

Path difference between trajectories 1 and 2:

ray 1 has longer distance to travel : $k\delta$

ray 2 moves slowly within the slab $\Rightarrow k(n-1)t$

$$\Rightarrow \quad k\delta = kt(n-1)$$

$$d \sin\theta = t(n-1)$$

But $\sin\theta = \dfrac{y'}{\sqrt{(y')^2 + L^2}} \approx \dfrac{y'}{L}$

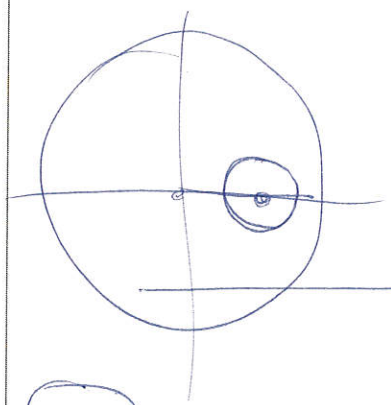
$$\Rightarrow y' = \frac{Lt(n-1)}{d}$$

$$E_y = \frac{\lambda}{4\pi\varepsilon_0 a} \frac{s}{\sqrt{1+s^2}} \Big|_0^{L} = \frac{\lambda}{4\pi\varepsilon_0 a}\left(\frac{L}{\sqrt{1+L^2}} - 0\right)$$

$$L \to \infty$$

$$\left(\frac{s}{\sqrt{1+s^2}}\right)' = \frac{1}{\sqrt{1+s^2}} - \frac{s}{(1+s^2)^{3/2}} \frac{1}{2}(2s) =$$

$$= \frac{1+s^2 - s^2}{(1+s^2)^{3/2}}$$



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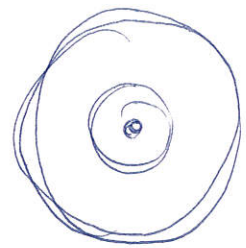
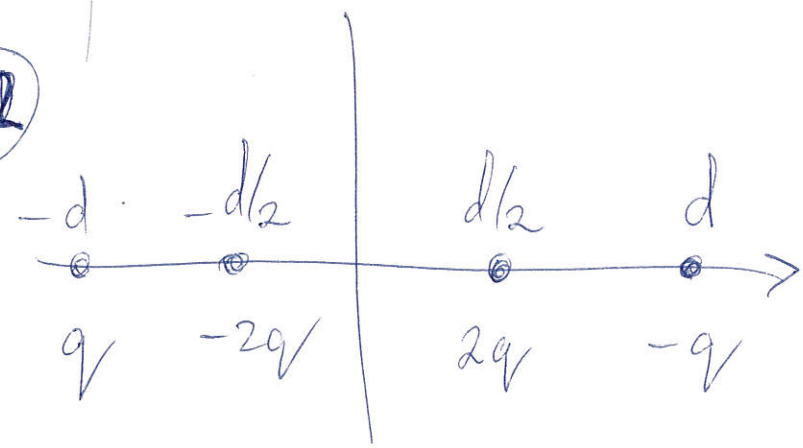


P2



5pts

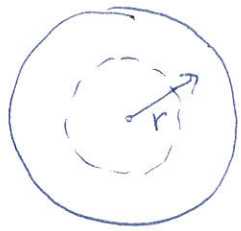
$$F = -\hat{x}\frac{2q^2}{(d/2)^2} - \hat{x}\frac{q^2}{(2d)^2} + \hat{x}\frac{2q^2}{\left(\frac{3}{2}d\right)^2}$$

$$= -\hat{x}\frac{q^2}{d^2}\left(8 + \frac{1}{4} - \frac{8}{9}\right) =$$

$$= -\frac{265}{36}\frac{q^2}{d^2}\hat{x}$$

(P3) Use superposition.

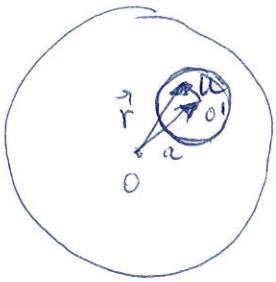
a)



El. field inside uniformly charged sphere

$$\vec{E}(r)\cdot 4\pi r^2 = \frac{\rho}{\varepsilon_0}\frac{4\pi}{3}r^3$$

$$\Rightarrow \vec{E}(r) = \frac{\rho\vec{r}}{3\varepsilon_0}$$



In our problem, $\vec{E}(r) = \vec{E}_1 + \vec{E}_2$

$\vec{E}_2$ = el. field of negatively charged sphere

$$\vec{E}_2(r) = \frac{-\rho\vec{u}}{3\varepsilon_0}$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = \frac{\rho\vec{r}}{3\varepsilon_0} - \frac{\rho\vec{u}}{3\varepsilon_0} = \frac{\rho(\vec{r}-\vec{u})}{3\varepsilon_0} = \frac{\rho\vec{a}}{3\varepsilon_0}$$

the field is uniform inside the hollow, and directed along vector $\vec{a}$.

b) $V_{O'} = V_1 + V_2$

V_1 = potential at point r inside uniformly charged sphere of charge density ρ.

$$\vec{E}_1 = \begin{cases} \frac{\rho R_1^3}{3\varepsilon_0}\frac{\hat{r}}{r^2}, & r > R \\ \frac{\rho}{3\varepsilon_0}\vec{r}, & r < R \end{cases}$$

$$V_1(r) = -\int_\infty^r \vec{E}\cdot d\vec{r} = -\int_\infty^{R_1} \frac{\rho R_1^3}{3\varepsilon_0}\frac{dr}{r^2} - \int_{R_1}^r \frac{\rho}{3\varepsilon_0} r\,dr =$$

$$= \frac{\rho R_1^3}{3\varepsilon_0} \frac{1}{r}\Big|_{\infty}^{R_1} - \frac{\rho}{3\varepsilon_0} \frac{1}{2} r^2 \Big|_{R_1}^{r} =$$

$$= \frac{\rho R_1^2}{3\varepsilon_0} - \frac{\rho}{6\varepsilon_0}\left(r^2 - R_1^2\right) = \frac{\rho}{6\varepsilon_0}\left(3R_1^2 - r^2\right)$$

$$\Rightarrow V_1 = \frac{\rho}{6\varepsilon_0}\left(3R_1^2 - r^2\right) \Rightarrow V_1(O') = \frac{\rho}{6\varepsilon_0}\left(3R_1^2 - a^2\right)$$

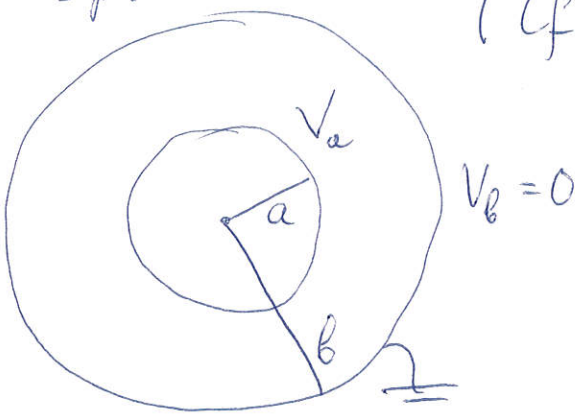
Hence $V_2 = \frac{-\rho}{6\varepsilon_0}\left(3R_2^2\right)$ potential of centre O' of the hollow (charge density $-\rho$)

$$\Rightarrow V_{O'} = \frac{\rho}{6\varepsilon_0}\left(3R_1^2 - a^2 - 3R_2^2\right)$$

$$= \frac{\rho}{6\varepsilon_0}\left[3\left(R_1^2 - R_2^2\right) - a^2\right].$$

(P4) 5pts

(Cf. ch. 3.2.2 of Griffiths)



$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) = 0 \Rightarrow r^2 \frac{\partial V}{\partial r} = C$$

$$\frac{dV}{dr} = \frac{C}{r^2}$$

$$\Rightarrow V(r) = A + \frac{B}{r}$$

BC: $\left.\begin{array}{l} V(a) = V_a \\ V(b) = 0 \end{array}\right\}$ $\left.\begin{array}{l} A + \frac{B}{a} = V_a \\ A + \frac{B}{b} = 0 \end{array}\right\}$

$\cancel{A = V_a}$

$$\left.\begin{array}{l} Aa + B = aV_a \\ Ab + B = 0 \end{array}\right\}$$

$$A(a-b) = aV_a$$

$$A = \frac{aV_a}{a-b}$$

$$B = -Ab = -\frac{abV_a}{a-b}$$

$$V(r) = \frac{aV_a}{a-b} - \frac{abV_a}{(a-b)r} = \frac{aV_a}{a-b}\left(1 - \frac{b}{r}\right)$$